A MAP FROM A FINITE MODULE TO THE LIMIT OF AN INVERSE SYSTEM INDUCING AN ISOMORPHISM ON ONE LOCAL COHOMOLOGY GROUP FORCES THAT SYSTEM TO BE ESSENTIALLY CONSTANT

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Editorial note. Statement span is lines 900-910 of sources/stacks-algebraization/pinned-original-source.tex; proof span is lines 912-924, the complete author proof (11 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Pool algebraization wave36 (al01 batch). Spans per computed label truth (al_clean_queue), not pool-card lines.

Editorial note. Cross-chapter homonym warning recorded in dedup (chapter-aware gate).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. See why-field; in-chapter lemmas cited by name inside the retained author proof become stub prereqs.

Source reference. Original lemma 0DX1: sources/stacks-algebraization/pinned-original-source.tex, source lines 881–893. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DX0: sources/stacks-algebraization/pinned-original-source.tex, source lines 819–830. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $(A, \mathfrak{m})$ be a Noetherian local ring. Let (M_n) be an inverse system of finite A-modules. Let $M \to \lim M_n$ be a map where M is a finite A-module such that for some i the map $H^i_{\mathfrak{m}}(M) \to \lim H^i_{\mathfrak{m}}(M_n)$ is an isomorphism. Then the inverse system $H^i_{\mathfrak{m}}(M_n)$ is essentially constant with value $H^i_{\mathfrak{m}}(M)$.*

Proof. By Lemma 0DX1 the inverse system $H^i_{\mathfrak{m}}(M_n)$ satisfies the Mittag-Leffler condition. Let $E_n \subset H^i_{\mathfrak{m}}(M_n)$ be the image of $H^i_{\mathfrak{m}}(M_{n'})$ for $n' \gg n$. Then (E_n) is an inverse system with surjective transition maps and $H^i_{\mathfrak{m}}(M) = \lim E_n$. Since $H^i_{\mathfrak{m}}(M)$ has the descending chain condition by Lemma 0DX0 we find there can only be a finite number of nontrivial kernels of the surjections $H^i_{\mathfrak{m}}(M) \to E_n$. Thus $E_n \to E_{n-1}$ is an isomorphism for all $n \gg 0$ as desired. □

Date: stacks-project:algebraization.tex:local-snapshot-20261012; GNU FDL 1.2 or later.